

Gilbert Uwonkunda

Senior Data Scientist · Geospatial AI Engineer

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PROFESSIONAL PROFILE

Senior Data Scientist and Geospatial AI Engineer with 5+ years designing and deploying production GeoAI and enterprise GIS systems for Rwanda's national government. Currently Senior Data Scientist on the World Bank SPARC (Spatial Planning and Resilience for Cities) Kigali Project, applying geospatial AI and data science to urban resilience, spatial planning, and city diagnostics. Architect of the Urban Dynamic Map for secondary and satellite cities at MININFRA; deployed a MaskRCNN illegal construction detection model for the City of Kigali; and built a real-time landslide risk scoring application serving field inspectors on web and mobile. Independently building ZoneAgent (zoneagent.live) — Rwanda's first LLM-powered land zoning tool. Trained and consulted for 20+ government institutions across East Africa.

CORE COMPETENCIES

GeoAI & Deep Learning	PyTorch, MaskRCNN, ResNet50, Vision Transformers, ArcGIS Deep Learning, Google Earth Engine, Computer Vision, Change Detection
Spatial Engineering	Python (ArcPy, GeoPandas, Rasterio), FastAPI, PostgreSQL/PostGIS, GDAL, REST APIs, Workflow Automation
Enterprise GIS	ArcGIS Enterprise, ArcGIS Pro, ArcGIS Mission, Survey123, SDE, Attribute Rules
Data & Analytics	Spatial Risk Modeling, Satellite Imagery Processing, Google BigQuery, SQL, Executive Dashboards
Domains	Urban Intelligence, Disaster Risk Reduction, Land Governance, Climate Monitoring, Government GIS

FLAGSHIP PROJECTS

ZoneAgent — Natural Language Land Zoning (Independent)

[Independent Project](#) · [GeoAI](#) · [zoneagent.live](#)

Independently building Rwanda's first LLM-powered land zoning assistant, enabling users to query land-use regulations through satellite imagery and natural-language conversation. Architecting the end-to-end product from data pipeline to user-facing web application.

Illegal Construction AI Detection — City of Kigali

[City of Kigali](#) · [ArcGIS Deep Learning](#)

Built and deployed a MaskRCNN instance-segmentation model (ResNet50 backbone) for the City of Kigali's AI-powered Building Inspection Hub — automatically flagging illegal construction and integrating with daily permit synchronization from BPMIS for live field-inspection workflows.

Urban Dynamic Map & Spatial Development Framework — MININFRA

[MININFRA](#) · [National Urban Planning](#)

Designed and deployed the Urban Dynamic Map and Spatial Development Framework for Rwanda's secondary and satellite cities — district-based planning dashboards, indicator reporting, and spatial data infrastructure supporting structured urban growth outside Kigali.

Landslide Risk Scoring Application — City of Kigali

[City of Kigali](#) · [Disaster Risk Reduction](#)

End-to-end system calculating per-building landslide exposure scores across Kigali, delivered via mobile (field) and web (management) platforms for disaster preparedness and urban resilience planning.

Forest Monitoring Deep Learning Model — Government of Rwanda

[National Forest Inventory](#) · [Climate & Land Use](#)

Custom deep learning model for automated forest change detection supporting Rwanda's National Forest Inventory and government climate-compliance reporting.

Rwanda Green Fund (GCF) Enterprise Platform

[Rwanda Green Fund](#) · [Climate Finance Tracking](#)

Architected and deployed a full ArcGIS Enterprise platform to track and report on internationally funded climate initiatives across Rwanda.

PROFESSIONAL EXPERIENCE

Senior Data Scientist

Apr 2026 – Present

World Bank — SPARC Kigali Project · Kigali, Rwanda

- Senior Data Scientist on the World Bank SPARC (Spatial Planning and Resilience for Cities) Kigali Project
- Applying geospatial AI and data-science methods to urban resilience, spatial planning, and city diagnostics for Kigali
- Working at the intersection of international development, urban data science, and applied GeoAI

Independent Builder & Lead Engineer

Jan 2026 – Present

ZoneAgent (zoneagent.live) · Kigali, Rwanda

- Independently building an AI-powered land governance and urban planning product for Africa
- Building ZoneAgent — a Natural Language Zoning tool enabling users to query land-use regulations through satellite imagery and conversational AI
- Architecting the end-to-end product from data pipeline to user-facing web application

Senior Geospatial AI Engineer

Jan 2025 – Apr 2026

Esri Rwanda · Kigali, Rwanda

- Lead architect for the City of Kigali AI-powered Building Inspection Hub — integrating NLA and CoK servers via VPN tunnels with automated daily permit synchronization from BPMIS
- Deployed a MaskRCNN illegal construction detection model (ResNet50 backbone) actively onboarded for field-inspection workflows
- Designed complex ArcGIS Enterprise attribute-rule workflows covering inspector approval chains, fine management, and automated case-status transitions
- Developed a Python-based REST API for daily permit reconciliation; produced executive dashboards and a parcel-permit mobile application for district leadership

GIS Specialist

Jun 2022 – Dec 2024

Esri Rwanda · Kigali, Rwanda

- Designed and deployed the Urban Dynamic Map (UDM) and Spatial Development Framework for Rwanda's secondary and satellite cities at MININFRA
- Built a landslide risk scoring application for the City of Kigali, calculating per-building risk scores delivered via mobile and web
- Trained a custom deep learning model for forest monitoring, enabling automated change detection for Rwanda's National Forest Inventory
- Architected the Rwanda Green Fund ArcGIS Enterprise platform for tracking internationally funded climate initiatives
- Deployed a real-time Health Emergency Management System using ArcGIS Mission, synchronizing ambulance response across Kigali
- Trained and consulted for 20+ government institutions across East Africa on enterprise GIS, spatial analysis, and digital transformation

GIS Consultant

Sept 2021 – Jun 2022

PSS Ltd · Kigali, Rwanda

- Delivered spatial analysis, GIS mapping, and custom geospatial workflow development for client projects across multiple sectors

SDG Coordinator

Aug 2020 – Jul 2021

UN SDSN Youth · Kigali, Rwanda

- Coordinated SDG youth-network programs and supported data collection, stakeholder engagement, and reporting for SDG-aligned initiatives in Rwanda

EDUCATION

Bachelor of Science in Geography
University of Rwanda — College of Science and Technology · Kigali, Rwanda

Sept 2016 – Aug 2021

CERTIFICATIONS

- Enterprise Geodata Management Professional 2201 · Esri · Jan 2025
- Generative AI Overview for Project Managers · PMI · Jun 2024
- ArcGIS Pro Associate 2101 · Esri · Jan 2023
- Google Project Management Certificate · Google · May 2023
- Esri Young Professionals Network (YPN) Ambassador · Esri · May 2024
- Google Advanced Data Analytics Certificate · Google · Jul 2023
- Databases and SQL for Data Science · IBM · Nov 2023
- Programming with JavaScript · Meta · Jul 2023